



PRAVEEN KUMAR OJHA

Process & Data Analyst

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MOTIVATION

Data Analyst with 4+ years of experience in Process Mining and Data Engineering at Bosch. Skilled in Celonis optimization, LLM-based conformance checking, and hands-on machine learning projects. Currently blending Analytics with process intelligence as part of my master thesis. Seeking roles where I can apply **data-driven thinking** to **optimize** and **automate** business processes.

 *Let's connect if you're solving complex process challenges or need sharp, scalable data analysis expertise!*

TECHNICAL SKILLS

CORE EXPERTISE

SQL		4/5
Machine Learning		4/5
Streamlit		4/5
Python		3/5
Git		4/5
PyCelonis		4/5
Prompt Engineering		4/5

TECHNOLOGY LANDSCAPE

Process Mining: Celonis, pm4py, OCPM, ETL

Optimization: Heuristic Opt., gurobipy, cplex

ML & Viz: scikit-learn, tensorflow, seaborn

Infra & Web: Flask, Docker, Grafana, Shell, pyodbc

PROCESS KNOWLEDGE

O2C (Order-to-Cash)

B2C (Bill-to-Cash)

P2P (Procure-to-Pay)

WORK EXPERIENCE

Master Thesis

 Bosch

 03/2025 - 07/2025

 Stuttgart, Germany

- Developing a modular Python application to benchmark over 10+ LLMs in parallel for BPMN conformance checking across 7 REST APIs.
- Optimizing workflows into DataFrames and integrating gold standard mapping evaluation, targeting 20% faster load times and enabling advanced conformance metrics like precision, recall, and F1 score.

Tech Stack:

Python

Streamlit

REST API

LLM

pm4py

pycelonis

Data Engineering in Process Mining Intern

 Bosch

 09/2024 - 02/2025

 Stuttgart, Germany

- Streamlined platform tasks (Data Pool backups, job health checks) with 60% faster execution, and automated regex-based outlier detection for access anomalies, cutting manual audit effort by 80%.
- Designed Streamlit application which automated process conformance checks; enabled structured activity mapping workflows used in 3+ pilot use cases during the PoC phase.
- Launched a Gamified leader board using LLM-scored commits to boost versioning, increasing developer engagement by 45% and automated weekly adoption tracking via Celonis ML Workbench and Action Flow.

Tech Stack:

Python

PyCelonis

SQL

Git / Gitflow

Streamlit

Working Student

 Bosch

 11/2023 - 08/2024

 Stuttgart, Germany

- Optimized Celonis transformation queries and data model loading, cutting use case runtime and memory usage by 40%.
- Built automated permission reporting dashboard, enabling quick detection of false permissions and improving reporting accuracy by 75%.

Tech Stack:

Python

Celonis

SQL

Action Flows

Senior Associate Consultant

 Bosch Global Software Solutions

 09/2019 - 08/2023

 Bangalore, India

- Led B2C process performance tuning, reducing runtime from 24 to 10 hours for 32M data through strategic query and memory optimization.
- Orchestrated process flow transition from Sales Order to Purchase Order, covering 3.5M records across 7 systems from a 13-month dataset.
- Supported AR use case setup, uploading a 30M-record data model from 7 systems via Oracle SQL to SAP HANA, enabling large-scale process analysis.

Tech Stack:

SQL

Celonis

B2C

P2P

I2P

A2A

IBC

ETL

Data Analyst Intern

 Tata Steel

 2018

 Jamshedpur, India

A2A (Affiliate-to-Business)

I2P (Invoice-to-Pay)

SOFT SKILLS

Agile Methodology

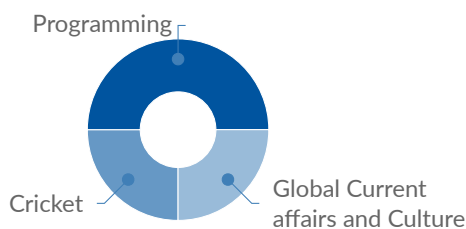
Process Consultation

Problem-Solving

LANGUAGES

Hindi Native
English C1 (Professional)
German A1 (Basic)

INTEREST AREAS



- **Developing Employee Turnover Prediction Model** - Developed a predictive model for employee turnover, identifying key attrition factors and helping HR design targeted retention strategies.

Tech Stack: Python Pandas Scikit-learn Matplotlib

EDUCATION

M.Sc. in Data Analytics and Decision Science

 **RWTH Aachen University**

 GPA - 1,5

 10/2023 – 09/2025

CO₂ EMISSIONS FORECASTING FOR EU COUNTRIES

ML ensemble prediction of EU carbon emissions

 Repository

- Developed a machine learning pipeline to **predict national CO₂ emissions** across EU countries by extending the methodology from a [2024 Scientific Reports paper](#).
- Benchmarked ML models (GBM, SVR, LSTM, Ensemble) across 10-year horizons in a Dockerized pipeline with Grafana dashboard and PostgreSQL-backed experimentation.

Tech Stack: Tensorflow Scikit-learn Seaborn Grafana Docker

OPTIMIZATION OF TRANSPORT PLANNING AND INBOUND PROCESSING IN PACKAGE CENTER

Routing and scheduling for overnight delivery

 Repository

- This project focuses on optimizing the parcel transportation network of DHL Deutsche Post, which involves overnight delivery across 37 centers.
- The optimization problem involves determining the departure times, routes, and loading strategies for swap bodies, while adhering to constraints such as sorting capacities, cut-off times, and driving regulations

Tech Stack: Python Gurobi Cplex Streamlit Git

BANK MARKETING CAMPAIGN

Machine Learning

 Repository

- The project focused on predicting customer subscription to "Term Deposit" offered by a Portuguese banking institution's marketing campaigns.
- Utilizing a dataset comprising 45,211 phone calls and 20 features, the problem was approached as a "Binary Classification" task, aiming to determine whether customers would opt for the offering, resulting in a Yes or No decision.

Tech Stack: Numpy Pandas Pytest Matplotlib

ECOFORCAST: REVOLUTIONIZING GREEN ENERGY SURPLUS PREDICTION IN EUROPE

Hackathon

- We were asked to develop a model to predict the country with the highest surplus of green energy in the upcoming hour among nine specified countries. Thus, considering renewable energy generation (e.g., wind, solar, geothermal) and energy consumption to calculate the surplus.
- Ranked 13th out of 237 teams in the hackathon challenge.

Tech Stack: Python Pandas NumPy scikit-learn Matplotlib

B.Tech. in Computer Science and Engineering

 **SOA University, India**

 CGPA: 9.83/10 (GPA: 1,1)

 07/2015 – 06/2019